

Министерство образования Республики Беларусь
Учреждение образования «Белорусский государственный университет
информатики и радиоэлектроники»

Факультет компьютерных систем и сетей
Кафедра Информатики
Дисциплина «Избранные главы информатики»

ОТЧЕТ
к лабораторной работе №1
на тему:
«СИСТЕМА КОНТРОЛЯ ВЕРСИЙ GIT»
БГУИР 6-05-0612-02

Выполнил студент группы 453503
ВОДНЕВ Кирилл Владимирович

(дата, подпись студента)

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(дата, подпись преподавателя)

Минск 2026

1. Подготовка рабочего окружения:

1. а. Установка Git:

```
PS C:\Users\kirya> git --version
git version 2.52.0.windows.1
```

1. б. Выполнение базовых настроек:

```
PS C:\Users\kirya> git config --global user.name vkwalker31
PS C:\Users\kirya> git config --global user.email kirya31pro@gmail.com
```

2. Создание структуры папок:

```
PS C:\Users\kirya> mkdir 453503_VODNEV_9

Каталог: C:\Users\kirya

Mode                LastWriteTime         Length Name
----                -
d-----          11.01.2026   15:27             453503_VODNEV_9

PS C:\Users\kirya> cd .\453503_VODNEV_9\
PS C:\Users\kirya\453503_VODNEV_9> git init
Initialized empty Git repository in C:/Users/kirya/453503_VODNEV_9/.git/
PS C:\Users\kirya\453503_VODNEV_9> git commit --allow-empty -m "Initialize main branch"
[master (root-commit) 518883f] Initialize main branch
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b IGI
Switched to a new branch 'IGI'
PS C:\Users\kirya\453503_VODNEV_9> mkdir IGI

Каталог: C:\Users\kirya\453503_VODNEV_9

Mode                LastWriteTime         Length Name
----                -
d-----          11.01.2026   15:28             IGI

PS C:\Users\kirya\453503_VODNEV_9> echo . > IGI\.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add IGI main directory"
[IGI e5ef971] Add IGI main directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 IGI/.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git checkout master
Switched to branch 'master'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b IGI_LR1 IGI
Switched to a new branch 'IGI_LR1'
PS C:\Users\kirya\453503_VODNEV_9> mkdir IGI_LR1

Каталог: C:\Users\kirya\453503_VODNEV_9\IGI

Mode                LastWriteTime         Length Name
----                -
d-----          11.01.2026   15:29             LR1

PS C:\Users\kirya\453503_VODNEV_9> echo . > .\IGI_LR1\.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git add .
```

```

PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add IGI/LR1 directory"
[IGI_LR1 9b44f19] Add IGI/LR1 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 IGI/LR1/.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI
Switched to branch 'IGI'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b IGI_LR2 IGI
Switched to a new branch 'IGI_LR2'
PS C:\Users\kirya\453503_VODNEV_9> mkdir IGI/LR2

```

Каталог: C:\Users\kirya\453503_VODNEV_9\IGI

Mode	LastWriteTime	Length	Name
d-----	11.01.2026 15:51		LR2

```

PS C:\Users\kirya\453503_VODNEV_9> echo . > .\IGI\LR2\.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add IGI/LR2 directory"
[IGI_LR2 057ad1a] Add IGI/LR2 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 IGI/LR2/.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI
Switched to branch 'IGI'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b IGI_LR3 IGI
Switched to a new branch 'IGI_LR3'
PS C:\Users\kirya\453503_VODNEV_9> mkdir IGI/LR3

```

Каталог: C:\Users\kirya\453503_VODNEV_9\IGI

Mode	LastWriteTime	Length	Name
d-----	11.01.2026 15:52		LR3

```

PS C:\Users\kirya\453503_VODNEV_9> echo . > .\IGI\LR3\.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add IGI/LR3 directory"
[IGI_LR3 60ba62d] Add IGI/LR3 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 IGI/LR3/.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI
Switched to branch 'IGI'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b IGI_LR4
Switched to a new branch 'IGI_LR4'
PS C:\Users\kirya\453503_VODNEV_9> mkdir IGI/LR4

```

Каталог: C:\Users\kirya\453503_VODNEV_9\IGI

Mode	LastWriteTime	Length	Name
d-----	11.01.2026 15:57		LR4

```

PS C:\Users\kirya\453503_VODNEV_9> echo . > .\IGI\LR4\.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add IGI/LR4 directory"
[IGI_LR4 82529d9] Add IGI/LR4 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 IGI/LR4/.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI
Switched to branch 'IGI'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b IGI_LR5
Switched to a new branch 'IGI_LR5'

```

```
PS C:\Users\kirya\453503_VODNEV_9> mkdir IGI/LR5
```

```
Каталог: C:\Users\kirya\453503_VODNEV_9\IGI
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	11.01.2026 15:57		LR5

```
PS C:\Users\kirya\453503_VODNEV_9> echo . > .\IGI\LR5\.gitkeep
```

```
PS C:\Users\kirya\453503_VODNEV_9> git add .
```

```
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add IGI/LR5 directory"
```

```
[IGI_LR5 f45df46] Add IGI/LR5 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 IGI/LR5/.gitkeep
```

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout master
```

```
Switched to branch 'master'
```

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b STRWEB master
```

```
Switched to a new branch 'STRWEB'
```

```
PS C:\Users\kirya\453503_VODNEV_9> mkdir STRWEB
```

```
Каталог: C:\Users\kirya\453503_VODNEV_9
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	11.01.2026 16:08		STRWEB

```
PS C:\Users\kirya\453503_VODNEV_9> echo . > .\STRWEB\.gitkeep
```

```
PS C:\Users\kirya\453503_VODNEV_9> git add .
```

```
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add STRWEB main directory"
```

```
[STRWEB 291b469] Add STRWEB main directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 STRWEB/.gitkeep
```

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout master
```

```
Switched to branch 'master'
```

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b STRWEB_LR1 STRWEB
```

```
Switched to a new branch 'STRWEB_LR1'
```

```
PS C:\Users\kirya\453503_VODNEV_9> mkdir STRWEB/LR1
```

```
Каталог: C:\Users\kirya\453503_VODNEV_9\STRWEB
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	11.01.2026 16:12		LR1

```
PS C:\Users\kirya\453503_VODNEV_9> echo . > .\STRWEB\LR1\.gitkeep
```

```
PS C:\Users\kirya\453503_VODNEV_9> git add .
```

```
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add STRWEB/LR1 directory"
```

```
[STRWEB_LR1 5bc13e7] Add STRWEB/LR1 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 STRWEB/LR1/.gitkeep
```

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout STRWEB
```

```
Switched to branch 'STRWEB'
```

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b STRWEB_LR2 STRWEB
```

```
Switched to a new branch 'STRWEB_LR2'
```

```
PS C:\Users\kirya\453503_VODNEV_9> mkdir STRWEB/LR2
```

```
Каталог: C:\Users\kirya\453503_VODNEV_9\STRWEB
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	11.01.2026 16:13		LR2


```

PS C:\Users\kirya\453503_VODNEV_9> echo . > .\STRWEB\LR2\.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add STRWEB/LR2 directory"
[STRWEB_LR2 8d704d6] Add STRWEB/LR2 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 STRWEB/LR2/.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git checkout STRWEB
Switched to branch 'STRWEB'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b STRWEB_LR3 STRWEB
Switched to a new branch 'STRWEB_LR3'
PS C:\Users\kirya\453503_VODNEV_9> mkdir STRWEB/LR3

```

Каталог: C:\Users\kirya\453503_VODNEV_9\STRWEB

Mode	LastWriteTime	Length	Name
d-----	11.01.2026 16:13		LR3

```

PS C:\Users\kirya\453503_VODNEV_9> echo . > .\STRWEB\LR3\.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add STRWEB/LR3 directory"
[STRWEB_LR3 02261d8] Add STRWEB/LR3 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 STRWEB/LR3/.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git checkout STRWEB
Switched to branch 'STRWEB'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b STRWEB_LR4 STRWEB
Switched to a new branch 'STRWEB_LR4'
PS C:\Users\kirya\453503_VODNEV_9> mkdir STRWEB/LR4

```

Каталог: C:\Users\kirya\453503_VODNEV_9\STRWEB

Mode	LastWriteTime	Length	Name
d-----	11.01.2026 16:14		LR4

```

PS C:\Users\kirya\453503_VODNEV_9> echo . > .\STRWEB\LR4\.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add STRWEB/LR4 directory"
[STRWEB_LR4 8897988] Add STRWEB/LR4 directory
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 STRWEB/LR4/.gitkeep
PS C:\Users\kirya\453503_VODNEV_9> git checkout STRWEB
Switched to branch 'STRWEB'

```

Создание структуры папок в соответствии с индивидуальным заданием:

```

PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI_LR1
Switched to branch 'IGI_LR1'
PS C:\Users\kirya\453503_VODNEV_9> mkdir .\IGI\LR1\bin\022
PS C:\Users\kirya\453503_VODNEV_9> mkdir .\IGI\LR1\bar\555
PS C:\Users\kirya\453503_VODNEV_9> mkdir .\IGI\LR1\baz\744
PS C:\Users\kirya\453503_VODNEV_9> echo 644 > .\IGI\LR1\baz\readme.txt
PS C:\Users\kirya\453503_VODNEV_9> mkdir .\IGI\LR1\baz\example\777
PS C:\Users\kirya\453503_VODNEV_9> mkdir .\IGI\LR1\foo\771
PS C:\Users\kirya\453503_VODNEV_9> mkdir .\IGI\LR1\foo\pwd\777
PS C:\Users\kirya\453503_VODNEV_9> echo 000 > .\IGI\LR1\foo\test.txt
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add IGI/LR1 directory structure"
[IGI_LR1 dc91b0c] Add IGI/LR1 directory structure
2 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 IGI/LR1/baz/readme.txt
create mode 100644 IGI/LR1/foo/test.txt

```

```
PS C:\Users\kirya\453503_VODNEV_9> tree /f
```

Структура папок

Серийный номер тома: CEC2-0C8B

C:.

```
├── IGI
│   ├── .gitkeep
│   └── LR1
│       ├── .gitkeep
│       ├── bar
│       │   └── 555
│       ├── baz
│       │   ├── readme.txt
│       │   ├── 744
│       │   ├── example
│       │   │   └── 777
│       ├── bin
│       │   └── 022
│       └── foo
│           ├── test.txt
│           ├── 771
│           └── pwd
│               └── 777
```

3. Клонирование проекта (https://github.com/smartiqaorg/geometric_lib) и удаление в нём .git для возможности создания коммита:

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI_LR1
```

Already on 'IGI_LR1'

```
PS C:\Users\kirya\453503_VODNEV_9> cd .\IGI\LR1\
```

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git clone https://github.com/smartiqaorg/geometric_lib.git
Cloning into 'geometric_lib'...
```

remote: Enumerating objects: 35, done.

remote: Total 35 (delta 0), reused 0 (delta 0), pack-reused 35 (from 1)

Receiving objects: 100% (35/35), 4.69 KiB | 436.00 KiB/s, done.

Resolving deltas: 100% (5/5), done.

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> Remove-Item -Recurse -Force .\geometric_lib\.git
```

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git add .
```

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git commit -m "Add cloned project"
```

[IGI_LR1 260a943] Add cloned project

3 files changed, 27 insertions(+)

create mode 100644 IGI/LR1/geometric_lib/circle.py

create mode 100644 IGI/LR1/geometric_lib/docs/README.md

create mode 100644 IGI/LR1/geometric_lib/square.py

4. а. Индексация нескольких файлов в Git-репозитории (git add ...):

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> echo 1 > 1.txt
```

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> echo 2 > 2.txt
```

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git add .\1.txt .\2.txt
```

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git status
```

On branch IGI_LR1

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

new file: 1.txt

new file: 2.txt

4. б. Внесение изменений в файлы:

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> echo 2 > 1.txt
```

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> echo 1 > 2.txt
```

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git status
```

```

On branch IGI_LR1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   1.txt
        new file:   2.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   1.txt
        modified:   2.txt

```

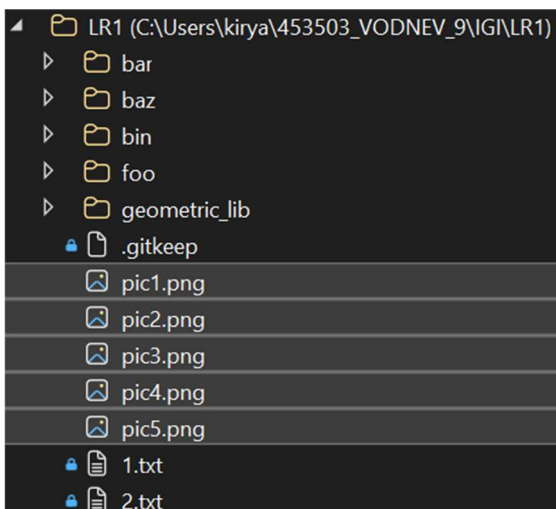
4. с. Выполнение коммитов к проиндексированным файлам:

```

PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git add .\1.txt .\2.txt
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git commit -m "Add two .txt files from Task 4.a - 4.c"
[IGI_LR1 c26d666] Add two .txt files from Task 4.a - 4.c
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 IGI/LR1/1.txt
 create mode 100644 IGI/LR1/2.txt

```

4. d. Добавление и удаление файлов (один или группу файлов с определенным расширением):



```

PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git add .\pic1.png .\pic2.png .\pic3.png .\pic4.png .\pic5.png
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git status
On branch IGI_LR1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   pic1.png
        new file:   pic2.png
        new file:   pic3.png
        new file:   pic4.png
        new file:   pic5.png

PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git rm pic1.png -f
rm 'IGI/LR1/pic1.png'
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git rm -r "*.png" -f
rm 'IGI/LR1/pic2.png'
rm 'IGI/LR1/pic3.png'
rm 'IGI/LR1/pic4.png'
rm 'IGI/LR1/pic5.png'
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git status
On branch IGI_LR1
nothing to commit, working tree clean

```

4. е. Переименование файлов:

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git mv .\1.txt text1.txt
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git mv .\2.txt text2.txt
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git status
On branch IGI_LR1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    renamed:   1.txt -> text1.txt
    renamed:   2.txt -> text2.txt

PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git commit -m "Rename two .txt files for Task 4.e"
[IGI_LR1 5e9cb20] Rename two .txt files for Task 4.e
 2 files changed, 0 insertions(+), 0 deletions(-)
 rename IGI/LR1/{1.txt => text1.txt} (100%)
 rename IGI/LR1/{2.txt => text2.txt} (100%)
```

4. ф. Создание веток для отдельной ЛР (выполнено в п. 2).

4. г. История коммитов: полностью и с ограничением вывода – отбор только последних, до и после указанной даты, по автору файлов, коммитов:

Все коммиты для ветки IGI_LR1 (git log):

```
PS C:\Users\kirya\453503_VODNEV_9> git log
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date:   Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date:   Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c

commit 260a943dee008ef18f1ae310e9c57c3317e8a8e5
Author: vkwalker31 <kirya31pro@gmail.com>
Date:   Sun Jan 11 17:19:24 2026 +0300

    Add cloned project

commit dc91b0c98c3ab8c030cef96e1924e88be49f8794
Author: vkwalker31 <kirya31pro@gmail.com>
Date:   Sun Jan 11 16:55:06 2026 +0300

    Add IGI/LR1 directory structure

commit 9b44f19e3713e2a7c69adc9626fbe76cb1981407
Author: vkwalker31 <kirya31pro@gmail.com>
Date:   Sun Jan 11 15:30:08 2026 +0300

    Add IGI/LR1 directory

commit e5ef9712d44ce6112f854699d227f1420aa3c385 (IGI)
Author: vkwalker31 <kirya31pro@gmail.com>
Date:   Sun Jan 11 15:28:51 2026 +0300

    Add IGI main directory

commit 518883f1b7682fe57b2245677e741c852aae4973 (master)
Author: vkwalker31 <kirya31pro@gmail.com>
Date:   Sun Jan 11 15:27:54 2026 +0300

    Initialize main branch
```


-n (показывает последние n коммитов)

```
PS C:\Users\kirya\453503_VODNEV_9> git log -3
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c

commit 260a943dee008ef18f1ae310e9c57c3317e8a8e5
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:19:24 2026 +0300

    Add cloned project
```

--before или --until (показывает коммиты, созданные до указанного времени):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --before "2026-01-11 15:30"
commit e5ef9712d44ce6112f854699d227f1420aa3c385 (IGI)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 15:28:51 2026 +0300

    Add IGI main directory

commit 518883f1b7682fe57b2245677e741c852aae4973 (master)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 15:27:54 2026 +0300

    Initialize main branch
```

--since или --after (показывает коммиты созданные после указанного времени):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --since "2026-01-11 15:30"
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c
```

--grep (фильтр коммитов по строке):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --all --grep "LR1"
commit dc91b0c98c3ab8c030cef96e1924e88be49f8794
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 16:55:06 2026 +0300

    Add IGI/LR1 directory structure

commit 5bc13e7e03ee63ae293bdf492df2d2a922ef6399 (STRWEB_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 16:12:40 2026 +0300

    Add STRWEB/LR1 directory

commit 9b44f19e3713e2a7c69adc9626fbe76cb1981407
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 15:30:08 2026 +0300

    Add IGI/LR1 directory
```

--author (фильтр коммитов по автору):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --author="vkwalker31"
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c
```

--committer (фильтр коммитов по пользователю, кто применил коммит в репозиторий):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --committer="vkwalker31"
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c
```

4. h. Исследование флагов для git log: p, stat, shortstat, name-only, name-status, relative-date, pretty, graph

-p (показывает изменения в каждом коммите):

```
PS C:\Users\kirya\453503_VODNEV_9> git log -p
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

diff --git a/IGI/LR1/1.txt b/IGI/LR1/text1.txt
similarity index 100%
rename from IGI/LR1/1.txt
rename to IGI/LR1/text1.txt
diff --git a/IGI/LR1/2.txt b/IGI/LR1/text2.txt
similarity index 100%
rename from IGI/LR1/2.txt
rename to IGI/LR1/text2.txt

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c

diff --git a/IGI/LR1/1.txt b/IGI/LR1/1.txt
new file mode 100644
index 0000000..c7250cb
Binary files /dev/null and b/IGI/LR1/1.txt differ
diff --git a/IGI/LR1/2.txt b/IGI/LR1/2.txt
new file mode 100644
index 0000000..ca90535
Binary files /dev/null and b/IGI/LR1/2.txt differ
```

--stat (показывает статистику изменений):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --stat
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

    IGI/LR1/{1.txt => text1.txt} | Bin
    IGI/LR1/{2.txt => text2.txt} | Bin
    2 files changed, 0 insertions(+), 0 deletions(-)

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c

    IGI/LR1/1.txt | Bin 0 -> 8 bytes
    IGI/LR1/2.txt | Bin 0 -> 8 bytes
    2 files changed, 0 insertions(+), 0 deletions(-)
```

--shortstat (показывает только итоговую статистику изменений):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --shortstat
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

2 files changed, 0 insertions(+), 0 deletions(-)

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c

2 files changed, 0 insertions(+), 0 deletions(-)
```

--name-only (выводит только названия файлов, изменённых в каждом коммите):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --name-only
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

IGI/LR1/text1.txt
IGI/LR1/text2.txt

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c

IGI/LR1/1.txt
IGI/LR1/2.txt
```

--name-status (выводит названия файлов и тип изменений для каждого файла):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --name-status
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

R100    IGI/LR1/1.txt    IGI/LR1/text1.txt
R100    IGI/LR1/2.txt    IGI/LR1/text2.txt

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c

A       IGI/LR1/1.txt
A       IGI/LR1/2.txt
```


--relative-date (отображает дату в формате относительно текущего времени):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --relative-date
commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: 39 minutes ago

    Rename two .txt files for Task 4.e

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: 56 minutes ago

    Add two .txt files from Task 4.a - 4.c
```

--pretty (позволяет кастомизировать формат вывода коммитов [oneline, full или short]):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --pretty=oneline
5e9cb20526a0f4c1c58c833e30eaa39a8e236633 (HEAD -> IGI_LR1) Rename two .txt files for Task 4.e
c26d6660206f395d17337f502607ad1fbc945277 Add two .txt files from Task 4.a - 4.c
260a943dee008ef18f1ae310e9c57c3317e8a8e5 Add cloned project
dc91b0c98c3ab8c030cef96e1924e88be49f8794 Add IGI/LR1 directory structure
9b44f19e3713e2a7c69adc9626f76cb1981407 Add IGI/LR1 directory
e5ef9712d44ce6112f854699d227f1420aa3c385 (IGI) Add IGI main directory
518883f1b7682fe57b2245677e741c852aae4973 (master) Initialize main branch
```

--graph (выводит в консоли графическое представление дерева коммитов, показывающее связи между ветками):

```
PS C:\Users\kirya\453503_VODNEV_9> git log --all --graph --oneline --decorate
* 5e9cb20 (HEAD -> IGI_LR1) Rename two .txt files for Task 4.e
* c26d666 Add two .txt files from Task 4.a - 4.c
* 260a943 Add cloned project
* dc91b0c Add IGI/LR1 directory structure
* 9b44f19 Add IGI/LR1 directory
| * 8897988 (STRWEB_LR4) Add STRWEB/LR4 directory
| * 02261d8 (STRWEB_LR3) Add STRWEB/LR3 directory
| /
| * 8d704d6 (STRWEB_LR2) Add STRWEB/LR2 directory
| /
| * 5bc13e7 (STRWEB_LR1) Add STRWEB/LR1 directory
| /
| * 291b469 (STRWEB) Add STRWEB main directory
| * f45df46 (IGI_LR5) Add IGI/LR5 directory
| /
| * 82529d9 (IGI_LR4) Add IGI/LR4 directory
| /
| * 60ba62d (IGI_LR3) Add IGI/LR3 directory
| /
| * 057ad1a (IGI_LR2) Add IGI/LR2 directory
| /
| * e5ef971 (IGI) Add IGI main directory
| /
* 518883f (master) Initialize main branch
```

4. i. Создание файла test.txt и вставьте в него строчку 'номер варианта'.
создание коммита и его отмена:

```
PS C:\Users\kirya\453503_VODNEV_9> echo 9 > test.txt
PS C:\Users\kirya\453503_VODNEV_9> git add .
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Commit to cancel"
[IGI_LR1 5df693b] Commit to cancel
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 test.txt
PS C:\Users\kirya\453503_VODNEV_9> git reset --soft HEAD~1
PS C:\Users\kirya\453503_VODNEV_9> git log -1
commit 718d7208445171d5fd1770d45940d72d6c951df1 (HEAD -> IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 18:57:58 2026 +0300

Add directory structure for 4f
```

4. j. Индексация файла в Git-репозитории и её отмена:

```
PS C:\Users\kirya\453503_VODNEV_9> git status
On branch IGI_LR1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   test.txt

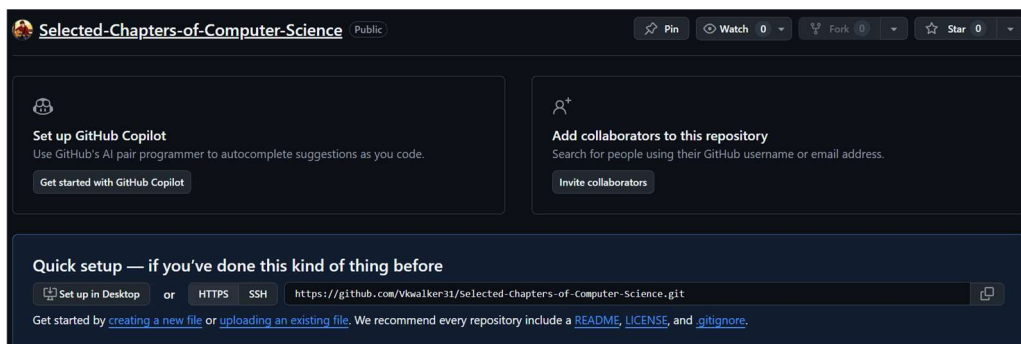
PS C:\Users\kirya\453503_VODNEV_9> git reset .\test.txt
PS C:\Users\kirya\453503_VODNEV_9> git status
On branch IGI_LR1
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test.txt

nothing added to commit but untracked files present (use "git add" to track)
```

4. k. Внесение изменений в файл в Git-репозитории и его возврат в исходное состояние:

```
PS C:\Users\kirya\453503_VODNEV_9> cd .\IGI\LR1\
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> cat .\text1.txt
2
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> echo 111 > .\text1.txt
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> cat .\text1.txt
111
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git checkout -- .\text1.txt
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> cat .\text1.txt
2
```

5. a. Создание удаленного репозитория на github.com:



5. b. Просмотр информации о нём и внесение изменений (remote, fetch):

```
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git remote add origin git@github.com:Vkwalker31/Selected-Chapters-of-Computer-Science.git
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git remote -v
origin  git@github.com:Vkwalker31/Selected-Chapters-of-Computer-Science.git (fetch)
origin  git@github.com:Vkwalker31/Selected-Chapters-of-Computer-Science.git (push)
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git remote set-url origin git@github.com:Vkwalker31/Selected-Chapters-of-Computer-Science.git
PS C:\Users\kirya\453503_VODNEV_9\IGI\LR1> git remote -v
origin  git@github.com:Vkwalker31/Selected-Chapters-of-Computer-Science.git (fetch)
origin  git@github.com:Vkwalker31/Selected-Chapters-of-Computer-Science.git (push)
```

5. c. Изменение нескольких файлов в проекте и перенесение их в удаленный репозиторий:

```
PS C:\Users\kirya\453503_VODNEV_9> git push --all origin
Enumerating objects: 62, done.
Counting objects: 100% (62/62), done.
Delta compression using up to 28 threads
Compressing objects: 100% (37/37), done.
Writing objects: 100% (62/62), 4.63 KiB | 474.00 KiB/s, done.
Total 62 (delta 13), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (13/13), done.
To https://github.com/Vkwalker31/Selected-Chapters-of-Computer-Science.git
* [new branch]      IGI -> IGI
* [new branch]      IGI_LR1 -> IGI_LR1
* [new branch]      IGI_LR2 -> IGI_LR2
* [new branch]      IGI_LR3 -> IGI_LR3
* [new branch]      IGI_LR4 -> IGI_LR4
* [new branch]      IGI_LR5 -> IGI_LR5
* [new branch]      STRWEB -> STRWEB
* [new branch]      STRWEB_LR1 -> STRWEB_LR1
* [new branch]      STRWEB_LR2 -> STRWEB_LR2
* [new branch]      STRWEB_LR3 -> STRWEB_LR3
* [new branch]      STRWEB_LR4 -> STRWEB_LR4
* [new branch]      master -> master
PS C:\Users\kirya\453503_VODNEV_9> git branch -r
origin/IGI
origin/IGI_LR1
origin/IGI_LR2
origin/IGI_LR3
origin/IGI_LR4
origin/IGI_LR5
origin/STRWEB
origin/STRWEB_LR1
origin/STRWEB_LR2
origin/STRWEB_LR3
origin/STRWEB_LR4
origin/master
```

5. d. Переименование удалённого репозитория:

Repository name

Selected-Chapters-Of-Computer-

Rename

Selected-Chapters-Of-Computer-Science is available.


```
PS C:\Users\kirya\453503_VODNEV_9> git remote -v
origin https://github.com/Vkwalker31/Selected-Chapters-of-Computer-Science.git (fetch)
origin https://github.com/Vkwalker31/Selected-Chapters-of-Computer-Science.git (push)
PS C:\Users\kirya\453503_VODNEV_9> git remote set-url origin https://github.com/Vkwalker31/Selected-Chapters-Of-Computer-Science.git
PS C:\Users\kirya\453503_VODNEV_9> git remote -v
origin https://github.com/Vkwalker31/Selected-Chapters-Of-Computer-Science.git (fetch)
origin https://github.com/Vkwalker31/Selected-Chapters-Of-Computer-Science.git (push)
```

5. е. Получение данных из удалённого репозитория

Эта команда скачает все ветки и обновления, но не объединит их с локальными ветками:

```
git fetch origin
```

Синхронизировать текущую ветку с удалённой:

```
git pull origin <название_ветки>
```

5. ф. Просмотр истории изменений

```
PS C:\Users\kirya\453503_VODNEV_9> git log origin/IGI_LR1
commit 718d7208445171d5fd1770d45940d72d6c951df1 (HEAD -> IGI_LR1, origin/IGI_LR1)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 18:57:58 2026 +0300

    Add directory structure for 4f

commit 5e9cb20526a0f4c1c58c833e30eaa39a8e236633
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:52:00 2026 +0300

    Rename two .txt files for Task 4.e

commit c26d6660206f395d17337f502607ad1fbc945277
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:34:14 2026 +0300

    Add two .txt files from Task 4.a - 4.c

commit 260a943dee008ef18f1ae310e9c57c3317e8a8e5
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 17:19:24 2026 +0300

    Add cloned project

commit dc91b0c98c3ab8c030cef96e1924e88be49f8794
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 16:55:06 2026 +0300

    Add IGI/LR1 directory structure

commit 9b44f19e3713e2a7c69adc9626fbe76cb1981407
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 15:30:08 2026 +0300

    Add IGI/LR1 directory

commit e5ef9712d44ce6112f854699d227f1420aa3c385 (origin/IGI, IGI)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 15:28:51 2026 +0300

    Add IGI main directory

commit 518883f1b7682fe57b2245677e741c852aae4973 (origin/master, master)
Author: vkwalker31 <kirya31pro@gmail.com>
Date: Sun Jan 11 15:27:54 2026 +0300

    Initialize main branch
```


5. г. Создание дополнительной ветки 1 в удалённом репозитории, перенос в неё двух файлов из другой ветки, удаление одного из файлов:

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI_LR1
Already on 'IGI_LR1'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b 1
Switched to a new branch '1'
PS C:\Users\kirya\453503_VODNEV_9> git add .\IGI\LR1\text1.txt .\IGI\LR1\text2.txt
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add two .txt files from IGI_LR1 to 1"
[1 dc9abad] Add two .txt files from IGI_LR1 to 1
1 file changed, 0 insertions(+), 0 deletions(-)
PS C:\Users\kirya\453503_VODNEV_9> rm .\IGI\LR1\text2.txt
PS C:\Users\kirya\453503_VODNEV_9> git add .\IGI\LR1\text2.txt
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Delete text2.txt"
[1 7b797ef] Delete text2.txt
1 file changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 IGI/LR1/text2.txt
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI_LR1
Switched to branch 'IGI_LR1'
PS C:\Users\kirya\453503_VODNEV_9> git branch -D 1
Deleted branch 1 (was 7b797ef).
```

5. h. Создание дополнительной ветки 2 в удаленном репозитории, перенос в неё двух файлов из проекта, слияние изменения в основную ветку, удаление дополнительной ветки 2:

```
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI_LR1
Already on 'IGI_LR1'
PS C:\Users\kirya\453503_VODNEV_9> git checkout -b 2
Switched to a new branch '2'
PS C:\Users\kirya\453503_VODNEV_9> git add .\IGI\LR1\text1.txt .\IGI\LR1\text2.txt
PS C:\Users\kirya\453503_VODNEV_9> git commit -m "Add two .txt files on branch 1"
On branch 2
nothing to commit, working tree clean
PS C:\Users\kirya\453503_VODNEV_9> git push origin 2
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   https://github.com/Vkwalker31/Selected-Chapters-of-Computer-Science.git
remote:
remote: Create a pull request for '2' on GitHub by visiting:
remote:   https://github.com/Vkwalker31/Selected-Chapters-of-Computer-Science/pull/new/2
remote:
To https://github.com/Vkwalker31/Selected-Chapters-Of-Computer-Science.git
 * [new branch]      2 -> 2
PS C:\Users\kirya\453503_VODNEV_9> git checkout IGI_LR1
Switched to branch 'IGI_LR1'
PS C:\Users\kirya\453503_VODNEV_9> git merge 2
Already up to date.
PS C:\Users\kirya\453503_VODNEV_9> git branch -d 2
Deleted branch 2 (was 718d720).
PS C:\Users\kirya\453503_VODNEV_9> git push origin --delete 2
remote: This repository moved. Please use the new location:
remote:   https://github.com/Vkwalker31/Selected-Chapters-of-Computer-Science.git
To https://github.com/Vkwalker31/Selected-Chapters-Of-Computer-Science.git
 - [deleted]          2
PS C:\Users\kirya\453503_VODNEV_9> git push origin IGI_LR1
Everything up-to-date
```